

## Communication

|                      |                                                             |
|----------------------|-------------------------------------------------------------|
| <b>Date</b>          | 6 July 2026                                                 |
| <b>Team ID</b>       |                                                             |
| <b>Project Name</b>  | OptiCrop: Smart Agricultural Production Optimization Engine |
| <b>Maximum Marks</b> | 1 Mark                                                      |

### Communication Plan:

Effective communication is essential for a successful project demonstration. This document outlines the communication strategy used within the team and with stakeholders throughout the project lifecycle, including how updates, issues, and feedback were managed.

| S.No | Communication Type      | Frequency       | Channel / Tool             | Participants             | Purpose                                                          |
|------|-------------------------|-----------------|----------------------------|--------------------------|------------------------------------------------------------------|
| 1    | Client Progress Reviews | Weekly          | Smartbridge Portal / Email | B Prudhvi Raj, Mentor    | To share progress on ML modeling and backend server integration. |
| 2    | Evaluation Checkpoints  | Milestone-based | GitHub / Portal            | B Prudhvi Raj, Evaluator | To verify features demo and model accuracy benchmarks.           |
| 3    | Self-Review & Coding    | Daily           | Local environment          | B Prudhvi Raj            | Coding functional components, UI screens, and testing routes.    |
| 4    | Stakeholder Review      | Bi-Weekly       |                            |                          |                                                                  |
| 5    | Final Demo Rehearsal    | Once            |                            |                          |                                                                  |
| 6    |                         |                 |                            |                          |                                                                  |

### Communication Challenges & Resolutions:

| S.No | Challenge Faced                                      | Resolution / Action Taken                                                                                |
|------|------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| 1    | Overlapping template images in Word conversions      | Programmed page breaks to isolate custom tables on new pages.                                            |
| 2    | ML model convergence warnings in Logistic Regression | Switched to Random Forest Classifier which achieved higher accuracy (99.32%) without convergence issues. |
| 3    | Interactive dashboard loading lag                    | Pre-compiled stats into crop_stats.json to load data instantly without re-querying dataframe.            |